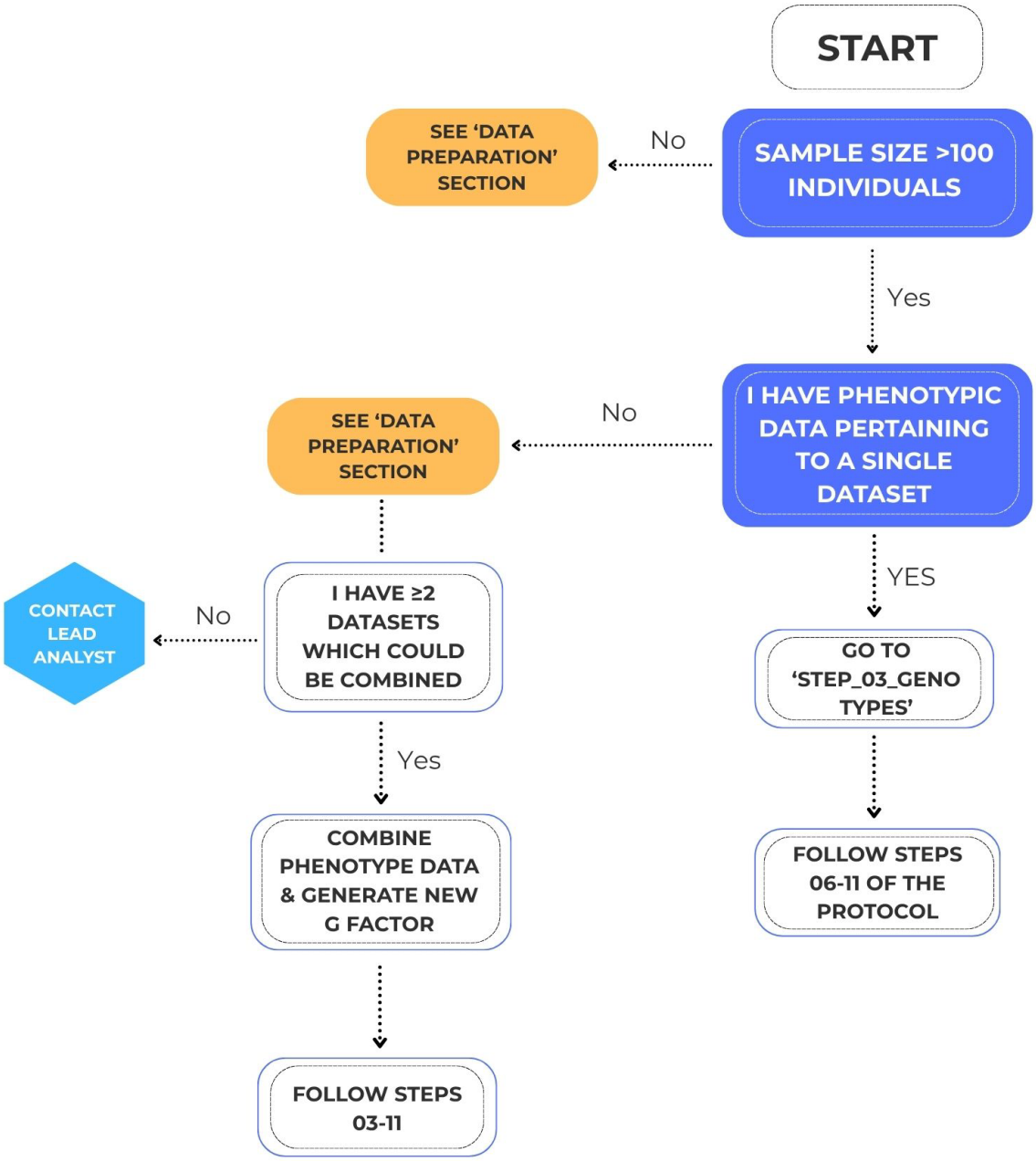
Exploring the Genetic Underpinnings of Cognition in Schizophrenia Case-only Samples: A Meta-analytic Approach

Genotype Protocol

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GENOTYPE PROTOCOL
FLOWCHART



PLEASE REFER TO THE **GENOTYPE PROTOCOL** FOR CONTEXT. IF UNCERTAIN, PLEASE CONTACT **LEAD ANALYST**, EVIE DOHERTY (E.DOHERY37@UNIVERSITYOFGALWAY.IE) FOR GUIDANCE

PGC Schizophrenia Cognition: Genotype Protocol

ID Matching

The complete-case samples from each dataset (as generated using the Phenotype Protocol: https://github.com/eviedoherty/PGC_SCZ_Cognition_Protocol/tree/main) should be matched to available PGC Wave 2, 3 and/or 4 genotype data on Snellius, the Dutch National Supercomputer.

Sample size: Each dataset should have >100 individuals after matching phenotype to genotype IDs.

Smaller datasets (<100): Datasets with <100 individuals will not be analysed in this iteration of the PGC SCZ – Cognition Subgroup analysis.

However, you may still be able to contribute by combining several smaller datasets from the same cohort or ancestry. Please see the **Data Preparation** section for more details.

Data Preparation

Datasets from the same data site which have a similar composition i.e. several of the same cognitive measures, with less than 100 samples should be collapsed and analysed together as a single unit (as per Nievergelt et al., 2024).

Before collapsing similar datasets, please ensure that the imputation methods are the same across datasets (differing imputation panels are acceptable between sites but not within sites). Note that after collapsing datasets, a '*g*' factor must be derived for the combined dataset before proceeding (Phenotype Protocol: https://github.com/eviedoherty/PGC_SCZ_Cognition_Protocol/tree/main).

If collapsing is not an option or you are unsure, please contact e.doherty37@universityofgalway.ie for guidance.

Genotypes

Both binary hard call genotype files (.bed, .bim, .fam) and dosage marker call (.fam, .gz, .info, .map, .ngt) files are required for this analysis.

Note: The dosage marker file extensions listed above are consistent with those used in PGC SCZ. However, dosage marker files may be available in alternative formats and therefore have different file extensions.

Hard Calls

For the Principal Component Analysis (PCA), the imputed binary genotype files i.e. hard calls, for the autosomal variants should be used to QC for high quality, high frequency variants (See **Principal Component Estimation**).

Dosage Marker Calls

Association testing should be performed using imputed dosage marker calls for all 22 autosomal chromosomes *and* the X chromosome. Dosage genotypes imputed via RICOPILI are output in PLINK2-dosage format, which is not compatible with GCTA fastGWA (Jiang et al., 2019). Files should therefore be converted to BGEN format using PLINK2 (see **01_convert_dosage_to_bgen_universal.sh**) and then combined per chromosome (**02_dosage_combine_per_chr.sh**).

Information: Light filtering should be applied to dosage marker calls prior to running the association analysis (MAF>0.005 and INFO>0.1). Please note that this filtering is automatically applied in the RICOPILI pipeline and therefore does not need to be repeated if imputation was performed via RICOPILI.

Combining Datasets

Hard Calls

Step 1: Match phenotype IDs to genotype IDs using the available .fam files and generate a .txt file containing the matching IDs. Please note that this is a general guide, and individual approaches to ID matching is permitted.

Note: Anything in **red** should be adjusted to match your path/file names.

Scenario 1 - Phenotype IDs fully match genotype IDs in the IID/FID column

The below code matches the phenotype IDs to the genotype **IID** column (column 2 .fam file; $2).

If the phenotype IDs match the **FID** column (column 1 .fam file), adjust the code accordingly (adjust "**$2** in ids" to "**$1** in ids"):

```
awk 'NR==FNR {ids[$1]; next} $2 in ids {print $1, $2}' \
./Folder/maindataset_IDs_matching_with_phenotypedataset.txt \
./SCZ_Wave3/maindataset/imputed/hardcall_genotypes_hg38_1kg/scz_ma
indataset_eur_sr-qc.hg38.ch.fl.bgn.fam \
> genotype_phenotype_matched_IDs.txt
```

Scenario 2 - Phenotype IDs *partly* match genotype IDs in the IID/FID column

In some cases, phenotype IDs only partly match genotype IDs For example, phenotype ID: **Dub_PT-2BBQ**, FID in .fam file: cas_scz_xdubl_eur_sr_A6.0***Dub_PT-2BBQ**.

The below code matches these phenotype IDs to genotype IDs in the FID column (to match with IID column adjust "{split(**$1**,a,"*")" to "{split(**$2**,a,"*")"):

```
awk 'NR==FNR {ids[$1]; next} {split($1,a,"*"); if (a[2] in ids)
print $1,
$2}' ./Folder/maindataset_IDs_matching_with_phenotypedataset.txt .
/SCZ_Wave3/maindataset/imputed/hardcall_genotypes_hg38_1kg/scz_dat
```

```
aset_eur_sr-qc.hg38.ch.fl.bgn.fam >
genotype_phenotype_matched_IDs.txt
```

Note: In the above example, the code matches phenotype IDs to genotype IDs using the portion of the genotype ID that follows "*". This symbol/character may vary across datasets and therefore should be adjusted accordingly depending on the format of your genotype IDs (i.e. "_" rather than "*").

Step 2: Filter the hard-call genotype files (binary files) to retain only individuals with available phenotypic data (i.e. a *g*-factor value). This ensures that PCs are generated in the same population included in the downstream association analysis.

Repeat the below code for each dataset you wish to merge i.e. if you have three datasets you wish to merge, then repeat the below code for all three datasets:

```
plink2 \
  --
bfile ./SCZ_Wave3/maindataset/imputed/hardcall_genotypes_hg38_1kg/
scz_dataset_eur_sr-qc.hg38.ch.fl.bgn \
  --keep ./folder/genotype_phenotype_matched_IDs.txt \
  --make-bed \
  --out hardcall_file_matched_IDs_dataset
```

Step 3: Merge the datasets

First, generate a merge list:

```
cat > merge_list_name.txt << EOF
hardcall_file_matched_IDs_dataset_1
hardcall_file_matched_IDs_dataset_2
EOF
```

Merge all files into a single dataset:

```
plink --bfile hardcall_file_matched_IDs_dataset_3 --merge-list
merge_list_name.txt --make-bed --out
dataset_hardcalls_merged_for_PCE
```

Note: This is a PLINK 1.9 command.

Your hard calls are now ready for **Principal Component Estimation**.

Dosage Marker Calls

The **merge_all_dataset_chrs.sh** script should be applied to datasets that have undergone collapsing as per the **Data Preparation** section. This will allow you to concatenate the dosage marker calls for the collapsed datasets. The outputs from the **dosage_combine_per_chr.sh** script should be used as inputs for the **merge_all_dataset_chrs.sh** script.

This script requires the use of QCTOOL (Band, 2021) which can be downloaded from: https://github.com/gavinband/qctool. This is not available as a module on Snellius and therefore will need to be loaded into your project from the GitHub repository.

Note: the concatenation function is not yet available through PLINK2 and when attempted with –pmerge, will flag the following error message: 'Non-concatenating --pmerge[-list] is under development', hence the use of QCTOOL.

Principal Component Estimation

Principal Component Estimation should be performed via the RICOPILI pipeline. Information on the installation and configuration of RICOPILI can be found at the following link: https://github.com/alicebraun/ricopili

Imputed variants with high imputation quality (INFO>0.80) and low missingness (<1%) should be carried forward for further quality control. Two iterations of linkage disequilibrium pruning (r^2=0.02, 200 SNP windows) and frequency filtering (MAF>0.05) should be applied to these high-quality SNPs for principal component estimation in each dataset. Long-range-LD areas should also be removed before performing PCA for each dataset (Chromosome 6: Major histocompatibility region (MHC) and Chromosome 8: Inversion region). A subset of the first 10 principal components generated for each dataset should be included in the final association analyses (see **Covariates**).

Note: All above parameters are automatically applied in RICOPILI pcaer. *Please note the number of remaining autosomal SNPs after applying this quality control step.*

Principal Component Estimation can be carried out using the pcaer function in RICOPILI. Before, running pcaer, symlink the directories containing the hard call files:

Note: Anything in **red** should be adjusted to match your path/file names.

Create symlinks

Step 1 - Create a new directory

```
mkdir pca_test
cd ./pca_test
```

Step 2 - Create the symlinks to the .bed, .bim, and .fam within that directory (required for pcaer)

```
ln -s
/home/your_name/SCZ_Wave3/dataset/imputed/hardcall_genotypes_hg38_
1kg/scz_dataset_eur_sr-qc.hg38.ch.fl.bgn.bim
ln -s
/home/your_name/SCZ_Wave3/dataset/imputed/hardcall_genotypes_hg38_
1kg/scz_dataset_eur_sr-qc.hg38.ch.fl.bgn.bed
ln -s
/home/your_name/SCZ_Wave3/dataset/imputed/hardcall_genotypes_hg38_
1kg/scz_dataset_eur_sr-qc.hg38.ch.fl.bgn.fam
```

Run pcaer

```
/path/to/RICOPILI/rp_bin/pcaer --prefercase --preferfam --out
pca_for_dataset scz_dataset_eur_sr-qc.hg38.ch.fl.bgn.bim
```

Output

The output file name should look like this; **mds_pca_for_dataset.menv**. This .menv folder will contain the PCs for your datasets (mds_pca_for_dataset.menv/pca_for_dataset.menv.emds) which should then be used in the association analysis as covariates.

Relatedness testing was not performed as the chosen association testing method (fastGWA) accounts for population stratification and relatedness using a genetic relationship matrix (see **Association Analysis**).

Main Association Analysis

Autosomal Variants

The association analysis for each dataset should be performed using the GCTA mixed linear model, fastGWA (Jiang et al., 2019). FastGWA uses a genetic relationship matrix (GRM) to account for population stratification and correct for genetic relatedness in the sample. Due to the continuous nature of '*g*', a linear model association analysis should be performed.

Download GCTA

The version of GCTA available on Snellius has a bug, hindering the generation of the GRM. Therefore, the newest version of GCTA was downloaded from the following link: https://yanglab.westlake.edu.cn/software/gcta/bin/gcta-1.95.3-linux-x86_64.zip.

Note: Both wget and curl failed to download GCTA directly on to Snellius as direct downloading from the Westlake University site (GCTA download site) to Snellius was blocked. Therefore, GCTA was downloaded to a local device as a .zip file and subsequently uploaded to Snellius for use.

Identify path once uploaded:

```
cd /home/your_directory_containing_gcta/gcta-1.95.3-linux-x86_64
./gcta --appimage-extract
ls squashfs-root/    #this should be present in the .zip file
ls squashfs-root/usr/bin/
```

To use:

The path identified using the code above should be used when running GCTA. It will likely resemble the example below, although the exact path may vary.

./squashfs-root/usr/bin/gcta64

Creating the GRM

The GRM should be generated for each dataset using a subset of high-quality variants with light pruning applied. Please note that the GRM was generated using the full sample available for each dataset to ensure it can be used for analyses beyond the current project.

The CEU HapMap3 variants were used to derive a subset of common and reliably genotyped variants. HapMap3 SNP lists can be downloaded from the following link: https://ftp.ncbi.nlm.nih.gov/hapmap/genotypes/hapmap3_r3/plink_format/hapmap3_r3_b36_fwd.qc.poly.tar.gz

Note: Anything in **red** should be adjusted to match your path/file names.

Step 1 - Generate a usable SNP list:

```
awk '{print $2}'
hapmap3_r2_b36_fwd.qc.poly/hapmap3_r3_b36_fwd.CEU.qc.poly.map >
hapmap3_CEU.snplist
```

Step 2 - Extract the high-quality variants from the merged dosage files per chromosome:

```
for chr in {1..22}
do
    plink2 \
        --bgen /path/to/per/chr/files/chr${chr}_merged.bgen ref-
first \
        --sample /path/to/per/chr/files/chr${chr}_merged.sample \
        --indep-pairwise 1000 100 0.9 \
        --extract /path/to/snplist/hapmap3_CEU.snplist \
        --rm-dup force-first \
        --out /path/for/output/file/chr${chr}_prune
```

```
done
```

Step 3 - Extract the pruned variants:

```
for chr in {1..22}
do
plink2 \
  --bgen /path/to/per/chr/files/chr${chr}_merged.bgen ref-first \
  --sample /path/to/per/chr/files/chr${chr}_merged.sample \
  --extract chr${chr}_prune.prune.in \
  --make-bed \
  --out chr${chr}_pruned_for_fastGWA
done
```

Step 4 - Merge the per chromosome pruned variants before generating the GRM partitions:

The following code uses PLINK 1.9 to merge the per chromosome pruned bfiles. PLINK 1.9 was uploaded to Snellius using wget and the following link https://s3.amazonaws.com/plink1-assets/plink_linux_x86_64_20260927.zip

Generate .txt file of bfiles to merge:

```
for chr in {1..22}; do
    echo "/path/to/pruned/files/chr${chr}_pruned_for_fastGWA"
done > /path/for/output/file/pruned_files_dataset.txt
```

Run merge:

```
/path/to/plink \ #plink 1.9
  --merge-list
/path/to/pruned/files/merge/list/pruned_files_dataset.txt \
  --make-bed \
  --out /path/to/combined/pruned/files/combined_pruned_for_fastGWA
```

Step 5 - Generate GRM partitions:

The GRM can be generated in several partitions using the **01_make_grm_partitions_fastgwa.sh** script.

To verify that the partitions have been generated successfully, run the following code to check that the output files contain data. An output of 0 indicates that the partitions have not been generated successfully:

```
ls -la grm_fastgwa_test.part_20_10.grm.N.bin
```

Note: By default, the script generates the GRM in 20 partitions. The number of partitions can be adjusted according to sample size, using approximately 1 partition per 100 individuals (e.g. 100 individuals = 1 partition; 250 individuals = 3 partitions).

Step 6 - Concatenate GRM partitions:

Once all the partitions have been created, they should be concatenated into a single file using the **02_concat_partitions_for_sparse_grm.sh** script.

Step 7 - Generate a Sparse GRM:

The sparse GRM can subsequently be generated using the **03_make_sparse_grm.sh** script.

Please ensure to adjust all scripts accordingly to match your data file names/paths.

Running GCTA fastGWA

The association analysis (fastGWA) should be run using the per chromosome BGEN files containing dosage marker calls (as generated by **02_dosage_combine_per_chr.sh** or **03_merge_all_dataset_chrs.sh** if the datasets have been collapsed).

Note: Before running the association analysis, please ensure that FID and IID columns match across the phenotype file (.tsv), chromosome sample file (.sample), and GRM ID file (.id). For example, the .sample file may only contain IDs in a single column, when IDs should be present in column 1 and column 2. This can be rectified using the following code on your .sample files:

```
for chr in {1..22}; do
awk 'BEGIN{OFS="\t"}
     NR==1{print "ID_1","ID_2","missing","sex"; next}
     NR==2{print "0","0","0","D"; next}
     {print $1,$1,$2,$3}' chr${chr}_merged.sample >
chr${chr}_merged_fixed.sample
done
```

Genotype Files

In fastGWA, each BGEN file must have a corresponding BGEN index file (.bgi). This can be generated in bgenix via Snellius using the following code:

Load and activate bgenix:

```
module load 2023
module load Anaconda3/2023.07-2
```

```
conda create -n bgen -c bioconda -c conda-forge bgenix
conda activate bgen
```

Find the bgenix path:

```
Which bgenix
```

Run from the path and generate index file to accompany .bgen file:

```
for chr in {1..22}; do
~/.conda/envs/bgen/bin/bgenix -g
/path/to/your/bgen_files/chr${chr}_merged.bgen   -index
done
```

Note: bgenix is not available as a module on Snellius, hence the use of a conda environment.

Phenotype File (.tsv)

Phenotype files should be in a tab separated format (.tsv). There should be three columns; FID, IID and the phenotype of interest which in this case is *g*.

FID and IID columns should match those in the chromosome sample (.sample), and GRM ID (.id) files.

Covariates

Each genetic analysis should be covaried for age, sex and a subset of the principal components derived in each cohort using the PCA (see **Principal Component Estimation**). The first four PCs should be included as covariates in all association

analyses, alongside any additional PCs that are nominally significantly associated with the *g*-factor phenotype ($P < 0.05$).

FastGWA requires continuous (--qcovar) and categorical (--covar) variables to be divided into separate files. Please ensure that covariate data is accompanied by identical FIDs and IIDs that match the .sample file belonging to the genotype files.

Covariates Files

Note: When generating covariates files, please ensures there are no NAs in the FID/IID columns as this will flag the following error:

Error: duplicated FID+IID in the covariate file.

This may occur if individuals have phenotype data and no genotype data.

Run fastGWA with covariate files:

```
/path/to/gcta-1.95.3-linux-x86_64/squashfs-root/usr/bin/gcta64 \
  --bgen /path/to/your/bgen_files/chr1_merged.bgen \
  --sample
/path/to/your/bgen_sample_files/chr1_merged_fixed.sample \
  --grm-sparse /path/to/your/sparse_grm/grm_fastgwa_sparse \
  --fastGWA-mlm \
  --pheno /path/to/your/phenotype_file/dataset_pheno_g.tsv \
  --qcovar /path/to/your/qcovar_file/dataset_qcovars.tsv \
  --covar /path/to/your/covar_file/dataset_covars.tsv \
  --thread-num 10 \
  --out fastGWA_c1_dataset
```

Note: This can be looped for efficiency.

Controlling for Inflated Test Statistics

To evaluate the effectiveness of quality control parameters on the genome-wide inflation of test statistics, a lambda genomic control (GC) calculation (Devlin & Roeder, 1999) should be performed. This should be performed on the combined summary statistics for a dataset (Chromosome 1-22). The following code can be used to combine summary statistic files:

```
head -n 1 /path/to/fastgwa_outputs/fastGWA_c1_dataset.fastGWA > Combined_g_GWAS_dataset_chr1_22.fastGWA
for i in {1..22} do
tail -n +2 /path/to/fastgwa_outputs/fastGWA_c${i}_dataset.fastGWA >> Combined_g_GWAS_dataset_chr1_22.fastGWA
done
```

The lambda GC value should fall within the desired range of 1.0-1.4 before the inclusion of principal components and 1.0-1.15 after the inclusion of principal components as covariates.

Although unlikely, in the case that the lambda GC value exceeds desired thresholds, quality control parameters should be adjusted i.e. made more stringent, until the value falls within normal range.

Note: As we are using a MLM (fastGWA), genomic inflation should already be accounted for through inclusion of the GRM.

Note: PGC dosage marker call files contain variants with an INFO score >0.1 and MAF >0.005, meaning these baseline parameters can be adjusted to ensure the lambda GC values falls within the range.

INFO scores and frequency metrics can be found in the corresponding .dosage.info files belonging to the original dosage marker outputs. A list of variants failing to exceed the adjusted thresholds can be derived from these files for extraction during re-analysis. Where the lambda GC falls outside the desired range, the association

analysis (fastGWA), should be re-run with the –extract function to remove the poor-quality variants.

Running Lambda GC in Snellius:

```
module load 2023
module load R/4.3.2-gfbf-2023a   # check exact version with:
module avail R
Rscript -e '
d <- read.table("Combined_g_GWAS_dataset_chr1_22.fastGWA",
header=TRUE)
chisq <- qchisq(d$P, df=1, lower.tail=FALSE)
lambda_gc <- median(chisq, na.rm=TRUE) / qchisq(0.5, df=1)
cat("Lambda GC:", lambda_gc, "\n")
'
```

Sex-stratified Association Analyses

The sex-stratified association analyses should be performed on males and females independently using the same approach outlined for the main association analyses (see **Main Association Analysis**).

Sex data can be acquired from the .sample/.fam file for each dataset, where 1=Male and 2=Female, or the covariate file containing sex (--covar). A list of individuals to keep should be generated from the .sample/.fam/covariates file and then included in the sex-specific analysis.

Generate the list of samples to keep:

Please note that this is a general guide, and individual approaches to generating a list of male/female IDs are permitted.

```
awk 'NR > 2 && $3 == "male" {print $1, $1}' OFS='\t'
covar_file_containing_sex_data.tsv >
/path/for/output/male_ids_dataset.txt
```

```
awk 'NR > 2 && $3 == "female" {print $1, $1}' OFS='\t'
covar_file_containing_sex_data.tsv >
/path/for/output/female_ids_dataset.txt
```

Run the sex-specific analyses:

```
/path/to/gcta-1.95.3-linux-x86_64/squashfs-root/usr/bin/gcta64 \
  --bgen /path/to/bgen_files/chr1_merged.bgen \
  --sample /path/to/bgen_sample_files/chr1_merged_fixed.sample \
  --grm-sparse /path/to/sparse_grm/grm_fastgwa_sparse \
  --fastGWA-mlm \
  --pheno /path/to/phenotype_file/dataset_pheno_g.tsv \
  --qcovar /path/to/qcovar_file/dataset_qcovars.tsv \
  --covar /path/to/covar_file/dataset_covars.tsv \
  --keep /path/to/list/of/IDs_to_keep/male_ids_dataset.txt
\   #adjust the –keep accordingly
  --thread-num 10 \
  --out /path/for/output/fastGWA_c1_male_only_dataset
```

Note: This can be looped for efficiency.

Chromosome X

The association analysis for chromosome X should be performed separately for male and female samples. Gene dosages in males should be scored 0 or 2 and in females as 0/1/2.

Note: This scoring has already been applied to the PGC SCZ chromosome X dosage files available on Snellius. Nevertheless, please double-check prior to running the analysis.

Due to the haploid nature of non-par regions in males, using .bgen formatted files with male X chromosome dosages will flag the following error in fastGWA:

Error: multiploidy detected in 0th SNP

As suggested by the developers of GCTA fastGWA, .pgen files should be used to run the X chromosome analysis in fastGWA as they can cope with the haploid/diploid disparity without flagging an error. Chromosome X file should be converted (**dosage_to_bgen_universal.sh**) and merged (**dosage_combine_per_chr.sh** and **merge_all_dataset_chrs.sh** (if applicable)) using the same approach as autosomes. To convert to .pgen format, the following additional step should be added:

```
plink2 \
  --bgen /path/to/bgen_files/edoherty/chrX_males.bgen ref-first \
  --sample /path/to/bgen_samples_files/chrX_males_fixed.sample \
  --make-pgen \
  --out chrX_males_only_dataset
```

Note: While the analysis will successfully run using .bgen format for female samples, to retain consistency, we ask that .pgen format is used for *both* males and females when running the association analysis on the X chromosome.

Running the X chromosome analyses:

```
/path/to/gcta-1.95.3-linux-x86_64/squashfs-root/usr/bin/gcta64
--pfile /path/to/pfile/chrX_males_only_dataset
--grm-sparse /path/to/sparse_grm/grm_fastgwa_sparse
--fastGWA-mlm
--dc 1
--pheno /path/to/phenotype_file/dataset_pheno_g.tsv
--qcovar /path/to/qcovar_file/dataset_qcovars.tsv
--thread-num 10
--out fastGWA_cX_males_dataset
```

Note: Sex should not be included as a covariate as the association analysis on the X chromosome is performed separately for males and females.